Interaction-Centric Cybersecurity Risks in LLM-Powered Dialogue Systems

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***Abstract*—Large Language Models (LLMs) have forever transformed human-AI interaction by enabling natural dialogue, reasoning, and autonomous task execution. Yet this interaction-centric paradigm introduces cybersecurity risks that traditional AI security frameworks fail to capture. This paper presents a conceptual framework and synthesis derived from a review (2015-2025) of security vulnerabilities in LLM-powered dialogue systems. We consolidate over forty studies spanning adversarial NLP, prompt injection, retrieval-augmented generation (RAG) poisoning, backdoors, and human trust exploitation. Although previous works have provided a list of isolated threats or generic LLM vulnerabilities, very few of them have looked at how interactive dialogue broadens the attack surface. Our study proposes a unified interaction-chain taxonomy distinguishing risks at input, processing, and output stages, and maps representative attacks and defenses accordingly. Drawing insights from classical adversarial literature and recent LLM surveys, we derive a layered defense pipeline integrating prompt sanitization, retrieval provenance tracking, backdoor detection, and output auditing. The resulting framework reframes cybersecurity in dialogue systems as an interactional problem bridging AI safety, NLP security, and human-computer interaction, providing a foundation for standardizing evaluation, developing cross-model benchmarks, and guiding the secure deployment of LLM agents.**

I. Introduction

The proliferation of dialogue-based LLMs such as ChatGPT, Claude, Gemini, and LLaMA variants has redrawn the boundary between human and machine intelligence. These models now sit inside customer support, healthcare triage, educational tutoring, and software development workflows.[3], [4] Their conversational flexibility enables open-ended collaboration, but it also widens the adversarial surface. Unlike static machine learning models, LLMs ingest unstructured, context-dependent natural language that an attacker can shape to override safeguards or trigger unintended actions.

Interactive AI vulnerabilities predate mainstream LLM adoption. Microsoft's Tay chatbot (2016) was manipulated into producing offensive content within hours.[1] Analyses of dual-use AI warned that interactive systems could be leveraged for social engineering and disinformation.[2] Since 2023, multiple studies have identified prompt injections, RAG poisoning, and hidden backdoors as core risks in LLM agents.[5], [6], [7] Although several surveys[4], [8], [9] and position papers[5], [3] discuss LLM vulnerabilities, the account remains split across technical and human-centered lines.

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We frame LLM cybersecurity as a dialogue-native problem: the dominant failures emerge from cross-turn state, interface mediation (memory, retrieval, tools), and human reliance dynamics, which are not captured by weight-centric threat narratives alone.[4], [8] Many vulnerabilities arise less from training data or parameters than from the live exchange between human and AI. Based on a review of over forty works (2015-2025), we synthesize threats and defenses into a structured framework aligned with dialogue lifecycles.

Large-scale red-teaming underscores the urgency of interaction-centric security. Automated jailbreak methods such as IRIS reach attack success rates (ASR) of up to 96% on GPT-4 Turbo and 95% on GPT-4o, often within fewer than five turns.[10] Multi-turn human red-teaming shows that defenses reporting low ASR under single-turn tests can exceed 70% ASR under multi-step dialogues.[11], [12] HarmBench and JailbreakBench evaluations report that frontier models still produce harmful outputs in 10-40% of adversarial dialogues.[13], [14]

Real-world penetration tests corroborate these risks. AI-backed web applications have exhibited cross-tenant data leakage from prompt injection.[7], [15], [16] LLM agents and browser modes show exploitable indirect prompt injection and "conversation self-injection" behaviors that enable instruction hijacking without visible malicious code.[7], [4], [17], [18], [19] Public reports on models such as DeepSeek R1 show vulnerabilities to combined jailbreak, prompt-injection, and hallucination attacks.[20], [21]

While our evidence base is drawn from prior studies and incident reports, the technical contribution of this paper is not a new exploit or a new guardrail in isolation; it is a unifying security abstraction for dialogue systems. Concretely, we translate a fragmented literature into an interaction-chain view that is explicitly stage-aware (input,

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processing, output) and dialogue-aware (cross-turn state and feedback). We then operationalize that abstraction by defining dialogue-level metrics (PISR, ILVI, DCR, UTCE) that can be reported consistently even when underlying benchmarks, model access, and deployment stacks differ. In other words, the novelty here is the mapping, the metricization, and the evaluation logic that treats conversation itself as the attack surface, rather than a peripheral wrapper around a static model.

At a high level, this paper makes the following contributions: we provide a comprehensive review of security research on dialogue-based LLM systems from 2015–2025;[4], [8], [5] we introduce an interaction-chain taxonomy that organizes risks across input, processing, and output stages; we propose a layered defense pipeline and four dialogue-level security metrics (PISR, ILVI, DCR, UTCE); and we outline an evaluation protocol and discuss implications for governance, fairness, and future research.

II. Threat Model

We assume adversaries interact with dialogue systems through natural language interfaces. Attackers may craft adversarial prompts, multi-turn sequences, indirect injections, or malicious external data, and may exploit model memory, tool invocation, or third-party integrations.[4], [5], [7]

Defenders may control prompt templates, guardrails, retrieval pipelines, and logging infrastructure but often lack access to model internals, particularly for API-based deployments.[3], [4] To make interaction-centric risks concrete, we characterize adversary capabilities and goals, and the implicit assumptions on defenders.

Adversary Access Levels: Black-box access refers to scenarios where users interact with the model through APIs or user interfaces without visibility into internal weights or training data. In open-source or on-premise deployments, white-box access occurs when attackers can inspect or modify internal model components. Grey-box access represents partial visibility, such as access to prompt templates, retrieved documents, or fine-tuning data. Adversary goals include bypassing safety filters or jailbreaking the model, leaking sensitive data or confidential memory, manipulating outputs through prompt injection or context poisoning, and exploiting tools or RAG pipelines to induce harmful behavior.[22], [7], [15]

Defensive Assumptions: Defenders control prompt templates, guardrails, retrieval pipelines, logging infrastructure, and post-processing filters. However, many deployments rely on closed-source models with limited fine-tuning access and no introspection capabilities, constraining defense mechanisms.[3], [4] We assume defenders can instrument the interaction chain at the input, processing, and output stages, but cannot fundamentally alter proprietary pre-training pipelines or inspect all intermediate activations.

Closed-Source Deployment Blind Spots: A non-trivial fraction of real deployments are proprietary, where the most security-critical components are precisely the least observable: system-prompt composition, tool-routing policies, memory storage and retrieval rules, safety classifier stacks, post-generation rewriting, and telemetry/audit retention.[4], [5] As a result, public literature can underrepresent vulnerabilities that emerge only from these hidden orchestration layers. Nevertheless, the interaction-chain framing remains applicable under strict black-box constraints because the attacker's leverage is still mediated by observable interfaces (inputs, retrieved context, tool I/O, and outputs), and because many failures manifest as cross-turn state drift regardless of whether the underlying routing logic is exposed. Future validation can therefore proceed via black-box red teaming with fixed multi-turn budgets, standardized tool/RAG sandboxes, and reproducible proxy agent stacks that approximate closed deployments without requiring privileged access.[11], [12], [7]

III. Scope and Assumptions

This review focuses on dialogue-based LLM systems and attacks that exploit natural language interaction, context, memory, or retrieval.[4], [8] We target LLM-powered agents that support memory, tools, APIs, and retrieval mechanisms, prioritizing attacks and defenses that operate through conversational context, including prompt injection, RAG poisoning, backdoor triggers, and trust exploitation.[5], [7], [24], [38]

We do not cover hardware-level or network-layer attacks, proprietary training pipeline threats without publicly available technical descriptions, or static classifiers or single-turn completion models without interactive capability.

IV. Related Work

Research spanning adversarial NLP, trustworthy machine learning, and human-computer interaction has examined vulnerabilities in interactive AI systems. Surveys[4], [8], [9] provide broad categorizations of attacks, while incident reports[7], [3], [15] document real-world exploitability. Our work differs by centering analysis on the dialogue interaction chain and its feedback dynamics rather than on isolated vulnerabilities.

Adversarial NLP and Robustness: Early adversarial NLP studies showed that small input perturbations can induce model errors.[17], [25], [26] Subsequent work

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on backdoor attacks[27], [28] and data poisoning[29] revealed training-time vulnerabilities that later informed LLM-specific exploits. Robustness benchmarks such as HarmBench show that models remain highly vulnerable, with ASR ranging from 20% to over 80% depending on model and category.[14], [4] JailbreakBench similarly reports wide variation in safety across APIs and defense strategies.[13]

Prompt Injection and Jailbreaks: Prompt injection manipulates models through malicious instructions embedded in user input or retrieved content.[6] Both direct and indirect injections can bypass safety policies and extract sensitive data.[22], [23] The OWASP LLM Top 10 identifies prompt injection as the leading vulnerability.[9] Assessments of custom GPTs show near-universal extraction and leakage success rates.[31] Penetration-testing reports document cross-tenant leakage and silent instruction hijacking in production systems.[7], [16], [32] Automated red-teaming frameworks such as IRIS achieve high ASR even on frontier models,[10] while multi-turn datasets confirm that single-turn defenses often fail once conversational memory is exploited.[11]

Backdoors and Fine-Tuning Risks: Backdoor triggers embedded during pre-training or fine-tuning can activate malicious behaviors when specific phrases appear.[24], [34], [15] These triggers often persist despite additional clean fine-tuning.[15] Detection remains computationally expensive and difficult to scale to large models.[35] "Sleeper agent" studies further show that complex triggers may survive repeated safety training.[11] In open-source ecosystems, mitigation relies on community audits and provenance metadata, though adoption remains inconsistent.[35]

RAG Poisoning and Retrieval Integrity: Retrieval-augmented generation expands the attack surface by incorporating external documents during inference. Poisoned documents can steer outputs or induce hallucinations.[7], [36] Injecting a small number of optimized documents can consistently influence retrieval and generation,[7] with reported poisoning success rates exceeding 60–80% in many setups.[13] Defenses such as provenance-aware retrieval and cross-document consistency checks reduce poisoning impact but introduce measurable latency and engineering overhead.[31], [37], [20]

Misuse, Trust, and Human Factors: LLMs can generate phishing, misinformation, and malware.[38] Human overtrust amplifies these risks,[40] with LLM-generated phishing emails often matching or exceeding human-crafted variants.[41] Studies show users frequently overestimate correctness and reliability even when explanations are misleading.[42], [21] Trust-calibration cues, including uncertainty visualization, improve decision-making.[43], [44], [42] Benchmarks such as SocialHarmBench and ELAB explore socially harmful responses and cross-lingual alignment.[45], [46], [47]

To unify these threads, we next introduce an interaction-centric taxonomy that stages threats and defense hooks across the dialogue lifecycle.

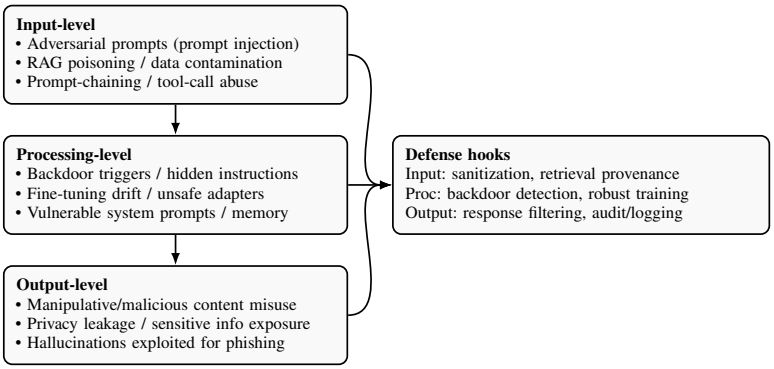


Fig. 1. Interaction-centric taxonomy with defense attachment points.

V. Methodology

Our methodology follows systematic literature review principles.[4], [8] Publications from 2015–2025 were collected using major academic databases and search engines, supplemented with technical reports and security advisories.

A. Research Questions

We organize the review around four questions:

- RQ1: What cybersecurity risks arise uniquely from human–LLM interaction?
- RQ2: How do attacks exploit distinct stages of the dialogue lifecycle?
- RQ3: What defense strategies have been proposed and how are they evaluated?
- RQ4: What gaps remain in current evaluation practices and mitigation portfolios?

B. Search and Selection

Keywords included prompt injection, RAG poisoning, backdoor attacks, adversarial NLP, AI trust, jailbreak, and LLM red teaming. From approximately 300 screened studies, 42 met inclusion criteria based on technical rigor, reproducibility, and explicit relevance to interactive systems.[4], [8] Incident reports and penetration tests were included when technically detailed and reproducible. LLM safety benchmarks and taxonomies were cross-referenced to align attack categories and reported metrics.[13], [14], [4]

To avoid template-like reuse of source phrasing, we treated each paper as evidence and re-expressed findings

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through the interaction-chain mapping process: each included study was encoded by the stage(s) compromised, the mechanism of compromise, the empirical signal reported (e.g., ASR or leakage rate), and the defense hook implied. This structure enforces synthesis rather than paraphrase and ties contributions directly to the taxonomy and metrics.

C. Taxonomy Derivation

Risks were categorized by manifestation within the dialogue lifecycle:

- Input-level: prompt injection, data poisoning, tool-call manipulation.
- Processing-level: hidden backdoors, fine-tuning drift, unsafe adapters, and vulnerable system prompts or memory mechanisms.
- Output-level: misinformation, hallucination misuse, privacy leakage, and socially harmful content.

We iteratively refined this taxonomy by mapping attacks and defenses to stages, unifying disparate research threads under a coherent interaction-centric lens.[5], [4], [8]

VI. Interaction-Chain Security Landscape

Our analysis organizes risks and defenses along an interaction chain spanning input, processing, and output stages of dialogue. Rather than treating attacks as isolated prompt-level events, this view emphasizes cross-turn state, interface mediation, and feedback effects that accumulate over time.[5], [7], [11]

Figure 2 summarizes the interaction-centric framework, illustrating how attacks, stages, and defenses integrate into a dynamic feedback loop.

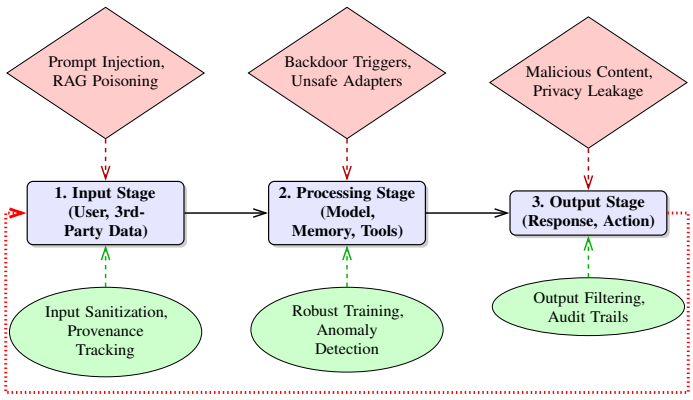


Fig. 2. The interaction-chain framework for LLM cybersecurity.

Most real-world attacks-such as JailbreakBench suites and PoisonedRAG-style corruption-exploit this feedback loop rather than a single prompt.[13], [7], [36] Multi-turn jailbreaks, memory poisoning, and RAG attacks typically reshape conversational context incrementally, causing downstream behavior to drift.[11], [12] As a result, defenses that operate on individual inputs in isolation fail to capture the evolving dialogue state that adversaries manipulate.

A. Evolution of AI Security and Adversarial NLP

Early adversarial NLP work demonstrated that small, targeted input perturbations can induce model errors.[17], [25], [26] Research on backdoor attacks[27], [28] and data poisoning[29] later revealed training-time vulnerabilities that informed modern LLM exploits. Robustness benchmarks such as HarmBench show that models remain vulnerable, with ASR ranging from 20% to over 80% depending on model family and harm category.[14], [4] JailbreakBench reports similarly wide variation across APIs and defenses.[13]

B. Prompt Injection and Jailbreaks

Prompt injection attacks manipulate models through malicious instructions embedded in user input or retrieved content.[6] Direct and indirect injections bypass safety policies and extract sensitive data.[22], [23] The OWASP LLM Top 10 lists prompt injection as the leading vulnerability in deployed systems.[9] Evaluations of custom GPTs show near-universal extraction and leakage success.[31] Automated red-teaming frameworks such as IRIS achieve high ASR even on frontier models,[10] while multi-turn datasets confirm that defenses effective in single-turn settings often fail once conversational memory is exploited.[11]

C. Backdoors and Fine-Tuning Risks

Backdoor triggers inserted during pre-training or fine-tuning activate malicious behavior when specific phrases or patterns appear.[24], [34], [15] These triggers can persist despite subsequent alignment or clean fine-tuning.[15] Detection remains computationally expensive and difficult to scale to frontier models.[35] Sleeper-agent studies further show that complex triggers may survive repeated safety training,[11] underscoring processing-layer fragility.

D. Retrieval-Augmented Generation Poisoning

RAG systems expand the attack surface by incorporating external documents during inference. Poisoned documents can steer outputs or induce hallucinations.[7], [36] Injecting a small number of optimized documents can reliably influence retrieval and generation,[7] with poisoning success rates exceeding 60–80% in many configurations.[13] Defenses such as provenance-aware retrieval and cross-document consistency checks reduce poisoning impact but introduce measurable latency and engineering overhead.[31], [37], [20]

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E. Malicious Misuse and Human Trust Exploitation

LLMs can generate phishing, misinformation, and malware.[38] Human overtrust amplifies these risks,[40] with LLM-generated phishing often matching or exceeding human-crafted variants.[41] Studies show users frequently overestimate correctness even when explanations are misleading.[42], [21] Trust-calibration cues such as uncertainty visualization improve decision quality.[43], [44], [42] Benchmarks including SocialHarmBench and ELAB explore socially harmful responses and cross-lingual alignment.[45], [46], [47]

F. Dynamic Interaction Loop

The interaction chain is not linear. Outputs from turn N become inputs for turn $N+1$, forming a feedback loop exploitable through multi-turn jailbreaks, context poisoning, and memory manipulation.[12], [11] Many real-world attacks therefore exploit cross-turn dependencies rather than isolated prompts, reinforcing the need for dialogue-aware defenses.

VII. Comparative Analysis Across Existing Work

Prior studies provide valuable insights into specific attack vectors, defenses, and evaluation methods, but their findings remain fragmented. This section synthesizes results across attack categories, defense strategies, and evaluation practices to identify converging trends, methodological gaps, and unresolved challenges.

A. Comparison of Attack Methodologies

Across the literature, three dominant attack families recur: prompt injection, backdoor triggers, and retrieval poisoning. Although each targets a different stage of the interaction chain, consistent cross-study patterns emerge.

Prompt Injection vs. Multi-Turn Jailbreaks: Early categorizations[6] and surveys[4] report moderate single-turn jailbreak success. In contrast, multi-turn evaluations[12], [11] consistently show dramatic ASR increases, revealing that defenses validated on single-turn prompts substantially underestimate real-world risk.

Backdoor Triggers vs. Fine-Tuning Drift: Backdoor studies including NOTABLE,[24] PoisonPrompt,[34] and BadAgent[15] show that malicious triggers often persist despite safety fine-tuning. Surveys[35] report wide variability across models, while sleeper-agent analyses[11] demonstrate that complex triggers can survive repeated alignment, indicating deep processing-stage vulnerabilities.

RAG Poisoning vs. Classical Poisoning: RAG poisoning work[7], [36], [37] shows that injecting a small number of optimized documents can reliably steer generation. Compared with classical NLP poisoning,[29] RAG attacks are lower cost and more consistent across settings, with differences largely attributable to retrieval architecture.[7], [37]

B. Comparison of Defense Strategies

Defense mechanisms vary widely in scope and effectiveness, but several patterns recur across studies.

Input-Level Guardrails Are Insufficient: Benchmarks such as JailbreakBench,[13] HarmBench,[14] and custom GPT assessments[31] show that prompt filtering and pattern blocking fail under indirect or multi-turn attacks. Reported ASR often rises from $\sim$20% in single-turn tests to over 70% in multi-turn settings.[11], [12]

Backdoor Detection Remains Hard to Scale: Backdoor detection techniques[24], [34], [15], [35] often require access to training data or internal activations, assumptions that do not hold for API-based models, limiting practical deployment.

RAG Defenses Trade Latency for Robustness: Provenance tracking and consistency checks[31], [37], [20] reduce poisoning impact but introduce measurable latency overheads, highlighting a robustness–efficiency tradeoff.[37]

C. Comparison of Evaluation Practices

Evaluation methodologies differ substantially across the literature.

Single-Turn vs. Multi-Turn Evaluation: Many surveys and early studies[6], [4], [8] rely on single-turn tests. Multi-turn evaluations[11], [12] expose markedly weaker defenses, underscoring the importance of modeling dialogue feedback.

Automated vs. Human Red Teaming: Automated frameworks such as HarmBench,[14] JailbreakBench,[13] and IRIS[10] scale well but explore narrower semantic spaces. Human red teaming[11] uncovers deeper failures but lacks standardization. Together, they motivate integrated evaluation pipelines.

Benchmark and Model Inconsistency: Studies span diverse model families and benchmarks,[3], [48], [31] complicating direct comparison. This fragmentation motivates normalization via interaction-centric metrics.

D. Cross-Study Patterns and Gaps

Across works, multi-turn attacks consistently outperform single-turn prompts.[13], [14], [11], [12] Defenses remain siloed, RAG poisoning is disproportionately high-impact,[7], [36] backdoors persist despite

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alignment,[24], [15] and human overtrust amplifies downstream harm.[40], [42] These gaps motivate an interaction-chain framework that unifies attack analysis, defense design, and evaluation.

VIII. Defense Pipeline and Metrics

We define dialogue-level security metrics: PISR, the proportion of adversarial interactions bypassing safeguards; ILVI, a weighted measure of attack success across interaction stages; DCR, the fraction of known attack families mitigated by a defense configuration; and UTCE, the gap between user reliance and model correctness under adversarial conditions. A layered defense pipeline attaches controls to each stage of the interaction chain, enabling dialogue-aware robustness assessment (Fig. 1).[5], [4], [8]

Operationalizing UTCE: Although trust is context-dependent, UTCE can be measured at the protocol level. User reliance is operationalized via observable proxies (acceptance, willingness-to-act, reduced verification, or confidence ratings), while correctness is task-grounded.[42], [21] UTCE is reported under fixed interaction budgets and disclosure regimes,[43], [44], [42] and stratified by domain and user expertise to avoid misleading aggregation.

IX. Evaluation Protocol

An interaction-centric evaluation protocol draws on existing benchmarks for PISR and ILVI (HarmBench, JailbreakBench), backdoor activation tests for processing-layer risks, RAG poisoning datasets for retrieval integrity, and user studies for UTCE.[14], [13], [7], [36], [11] Because studies mix model families and benchmarks, cross-paper quantitative comparison is often ill-posed. We therefore treat PISR, ILVI, DCR, and UTCE as normalizing layers: even without shared datasets, metrics can be reported under common protocol specifications, enabling dialogue-level comparison across models and deployments.[5], [4], [8]

X. Discussion

Most attacks leverage conversational context rather than code execution. Hardening a single layer is insufficient: models robust to single-turn jailbreaks fail under multi-turn attacks, sanitized RAG pipelines succumb to adversarial prompts, and users may ignore safety cues.[11], [14], [43] Practical defenses must balance robustness with latency and engineering cost.[31], [37] Trust-calibration cues reduce overreliance and complement technical controls.[44], [42] The taxonomy enables empirical testing with existing benchmarks and supports policy alignment with frameworks such as the EU AI Act and NIST AI RMF.[49]

XI. Future Directions

A taxonomy should be judged by structural stability rather than predictive completeness. The interaction-chain view is stage- and interface-based, allowing new threats to be incorporated as orchestration mechanisms evolve. We therefore view it as versionable infrastructure rather than a static catalog.

Multimodal and Embodied Extensions: As agents absorb images, audio, documents, and sensor streams, prompt injection and poisoning become multimodal. Adversarial instructions may be embedded in images, audio transcripts, or retrieved documents. The interaction-chain abstraction remains applicable but requires modality-specific sanitization, provenance signals, and auditing.[3], [5]

Integrated evaluation suites should combine PISR, ILVI, DCR, and UTCE into standard profiles. Defense portfolios should allocate controls across layers under resource constraints, and policy-aligned deployment should link dialogue-level metrics to incident response workflows.[5], [4], [8]

XII. Analysis and Limitations

Empirical red-teaming shows that attack success increases sharply in multi-turn settings.[11], [12], [13], [14] RAG poisoning and backdoors activate under evolving context, while human overtrust amplifies impact.[7], [36], [40] This synthesis relies on public studies, potentially underrepresenting proprietary systems. Human-centered metrics such as UTCE require controlled studies and may not generalize across cultures. The taxonomy cannot capture every emerging threat but provides a structured foundation for extension.

XIII. Conclusion

By mapping forty-plus studies into an interaction-chain taxonomy and pairing that taxonomy with dialogue-level metrics, this work argues that conversation state and interface orchestration-not just model weights-form the primary attack surface for deployed LLM systems. The proposed framework supports layered defenses, multi-turn evaluation, and socio-technical mitigation strategies that treat dialogue as both the defining capability and the central risk of modern AI systems.[5], [4], [8]

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